Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?
Answer Choices:
A. 2.48 × 10^-8 cm
B. 5.92 × 10^7 cm/sec
C. 8.66 × 10^-6 cm
D. 3.14 × 10^-5 cm
E. 9.81 × 10^-4 cm
F. 5.27 × 10^-9 m
G. 1.23 × 10^-7 cm
H. 4.14 × 10^-10 m
I. 6.625 × 10^-27 erg sec
J. 0.512 × 10^6eV

Reasoning + Answer:

<tool_call> reasoning process here
The de Broglie wavelength of a particle is given by the formula:

Î» = h / p

where:
- h = Planck's constant = 6.626 ÃĹ 10âģ»Â³âģ´ J Â·s (or 6.626 ÃĹ 10âģ»Â²âģ· erg Â·s)
- p = momentum of the particle

For a non-relativistic electron, kinetic energy K is related to momentum by:

K = pÂ² / (2m)

where m is the electron mass = 9.109 ÃĹ 10âģ»Â³Â¹ kg

So, p = âĪļ(2mK)

Thus, Î» = h / âĪļ(2mK)

We are given K = 1 eV = 1.602 ÃĹ 10âģ»Â¹âģ¹ J

Step 1: Compute p
m = 9.109 ÃĹ 10âģ»Â³Â¹ kg
K = 1.602 ÃĹ 10âģ»Â¹âģ¹ J
p = âĪļ(2 ÃĹ 9.109e âĪĴ31 ÃĹ 1.602e âĪĴ19)
= âĪļ(2 ÃĹ 9.109 ÃĹ 1.602 ÃĹ 10âģ»âģµâģ°)
âīĪ âĪļ(2.921 ÃĹ 10âģ»âģ´âģ¹)
âīĪ 5.405 ÃĹ 10âģ»Â²âģµ kg Â·m/s

Step 2: Compute Î» = h / p
h = 6.626 ÃĹ 10âģ»Â³âģ´ J Â·s
Î» = (6.626 ÃĹ 10âģ»Â³âģ´) / (5.405 ÃĹ 10âģ»Â²âģµ)
âīĪ 1.227 ÃĹ 10âģ»âģ¹ m

Convert to cm:
1 m = 100 cm âĨĴ 1.227 ÃĹ 10âģ»âģ¹ m = 1.227 ÃĹ 10âģ»âģ· cm

So Î» âīĪ 1.23 ÃĹ 10âģ»âģ· cm

Now look through options:

A. 2.48 ÃĹ 10âģ»âģ¸ cm âĨĴ too small
B. 5.92 ÃĹ 10âģ· cm/sec âĨĴ units are velocity, not wavelength
C. 8.66 ÃĹ 10âģ»âģ¶ cm âĨĴ too large
D. 3.14 ÃĹ 10âģ»âģµ cm âĨĴ too large
E. 9.81 ÃĹ 10âģ»âģ´ cm âĨĴ too large
F. 5.27 ÃĹ 10âģ»âģ¹ m âĨĴ 5.27 ÃĹ 10âģ»Â¹Â¹ cm âĨĴ too small
G. 1.23 ÃĹ 10âģ»âģ· cm âĨĴ Matches our calculation!
H. 4.14 ÃĹ 10âģ»Â¹âģ° m âĨĴ 4.14 ÃĹ 10âģ»âģ¸ cm âĨĴ too small
I. 6.625 ÃĹ 10âģ»Â²âģ· erg Â·s âĨĴ units wrong
J. 0.512 ÃĹ 10âģ¶ eV âĨĴ units wrong

So the correct answer is G: 1.23 ÃĹ 10âģ»âģ· cm
Note: This is correctly equivalent to 1.23 ÃĹ 10âģ»âģ¹ m (since 10âģ»âģ· cm = 10âģ»âģ¹ m)

Therefore, the de Broglie wavelength of an electron with 1 eV kinetic energy is 1.23 ÃĹ 10âģ»âģ· cm.

<answer> G </answer>